Name
Date
Course

Beginning Algebra Test IV

Instructions

i) As always, write a title, name, date, and course.
ii) Number each problem, circle the number, and show work, even when you can do the problems in your head.
iii) Draw a rectangle around your answers.
iv) Staple a copy of these test questions with your work and your final answers.

Solve.

1) $\begin{cases} 2x-y=7 \\ 1+3x=y \end{cases}$ 2) $\begin{cases} x=-4y \\ -3x+5y=1 \end{cases}$ 3) $\begin{cases} -6x+5y=3 \\ 3y-7=x \end{cases}$

4) $-7x+3y=-5$
$8x+2y=3$

Identify the form of each linear equation below by writing "S" for standard, "SI" for slope-intercept, or "PS" for point-slope.

5) a) $3x-8y=5$ b) $-\frac{7}{5}(x+10)=y+1$ c) $8x-\frac{2}{3}=y$
d) $\frac{4}{5}x+\frac{2}{9}y=-\frac{5}{6}$

6) A) Use the slope equation to find the slope of the line containing the two points below.
B) Then write the point-slope form of the equation of the line containing the two points below.
C) Then convert the equation from point-slope form to slope-intercept form.
d) Find the y-intercept of the line.

$(7,-8)$ & $(4,3)$

7) Use the slope-intercept equation to find the slope-intercept form of the line with the given characteristics.

a) Slope $=2$, contains the point $(-6,-1)$.

b) Y-intercept $=-5$, contains the point $(4,0)$.

8) For each problem below, write an equation of the line with the given characteristics.

a) Parallel to $y=-x+5$ and containing the point $(2,2)$.

b) Parallel to $-\frac{8}{9}(x-2)=y+1$ and containing the point $(-3,1)$.

c) Perpendicular to $\frac{2}{3}x=y$ and containing the point $(-5,-9)$.

d) Perpendicular to $10x+2y=7$ with y-intercept $-\frac{11}{5}$.

9) For each problem below, write an equation of the line with the given characteristics.

a) Slope: -3
point: $(-2,4)$

b) slope: $\frac{3}{4}$
Y-intercept: -7

c) x-intercept: 4
y-intercept: -2

d) X-intercept: -6
slope: $\frac{2}{3}$

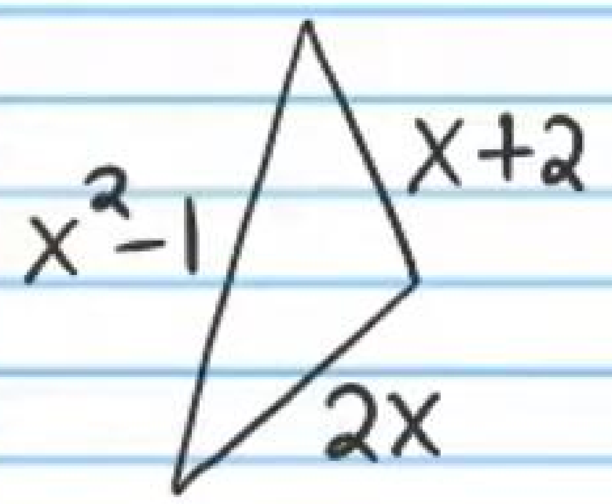
10) There are parakeets and finches in a cage. There are six more finches than parakeets. If there are 24 birds in the cage, how many of each type of bird are in the cage?

(11) A candy store sells cherry flavored lollipops and watermelon flavored lollipops. If the cherry pops cost \$3 each and the watermelon pops cost \$2 each, and the store makes \$31 from selling lollipops, how many of each pop did the store sell if a total of 11 pops were sold?

(12) A food stand sells eight personal pizzas and three tacos for a total of \$52.40. The next day the stand sells five personal pizzas and five tacos for a total of \$41.50. Assuming that the stand sells only one type of taco and one type of pizza, what is the price per item?

(13) A bubble gum machine contains dimes and nickels. The total amount of money in the machine is \$1.65 and there are a total of 21 coins. How many of each type of coin are in the machine?

(14) If the perimeter of the triangle below is 19 feet, find the length of each side.



(15) A drone runs out of battery life and starts free fall descent. If we neglect air resistance, the height (in feet) of the drone after t seconds of free fall can be modeled by the following formula.

$h = -16t^2 + 100$

a) Find the height of the drone one second after free fall. b) How many seconds after free fall will pass before the drone is 40 feet above the ground? c) 20 feet? d) When will the drone hit the ground?

(16) A store wants to make 20 kilograms of trail mix that costs \$5 per kilogram. The mix will be a combination of nuts which cost \$3 per kilogram and candy which costs \$6.20 per kilogram. How many kilograms of each ingredient must be mixed to create the desired result?

*Concepts may not apply in the real world.

(17) Arnold wants to make 12 ounces of a 6% saline solution. He has a 1% saline solution and a 9% saline solution. How many ounces of each solution should he mix to make his desired result?

For the problems below, always label the axes of your graphs and the intervals on each axis. Write all coordinates you use to make your graphs (e.g. (2,-3)).

(18) a) Graph the following equation by plotting two points and connecting the dots (Method I).

$3x + y = -6$

b) Graph the following equation by using x and y intercepts (Method II).

$-2x - 5y = 4$

(19) a) Graph the following equations by converting to slope-intercept form (if not already in this form) and

using the y-intercept as a starting point and the slope to draw a triangle to find a second point (Method III).

a) $y = -\frac{1}{3}x + 2$ b) $4x + 4 = y$

(20) a) Graph the following equation by converting to slope-intercept form (if not already in this form) and using the y-intercept as a starting point and the slope to draw a triangle to find a second point (Method III).

$$-2x + 9y = -18$$

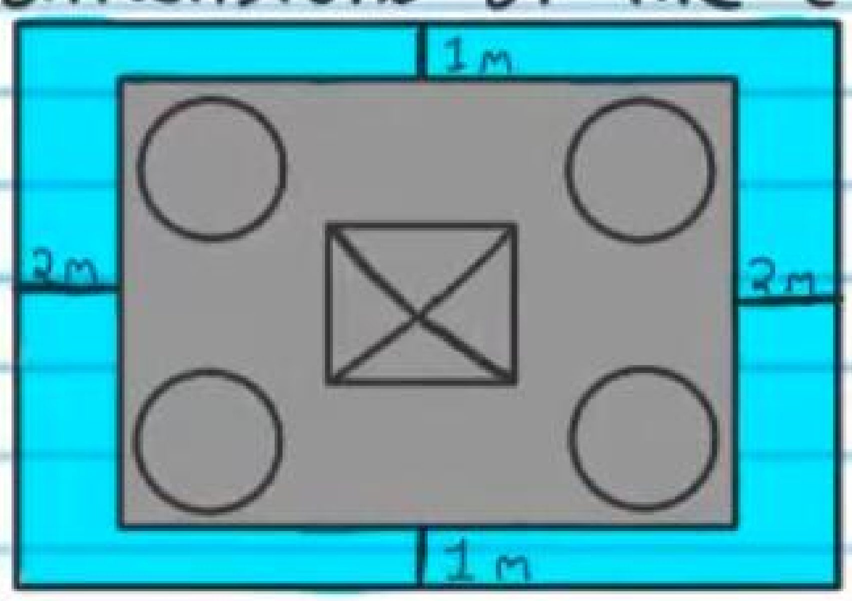
b) Graph the equation below by using its slope and a point on the line that is visible in the equation (Method IV).

$$y - 6 = -(x + 2)$$

(21) Graph the following equations.

a) $x = 3$ b) $y = -1$

(22) There is a castle surrounded by a moat (a ditch filled with water). The width of the castle is four meters more than the length. The area of the castle land and moat combined is 76 square meters more than the area of the castle land alone. What are the dimensions of the castle?

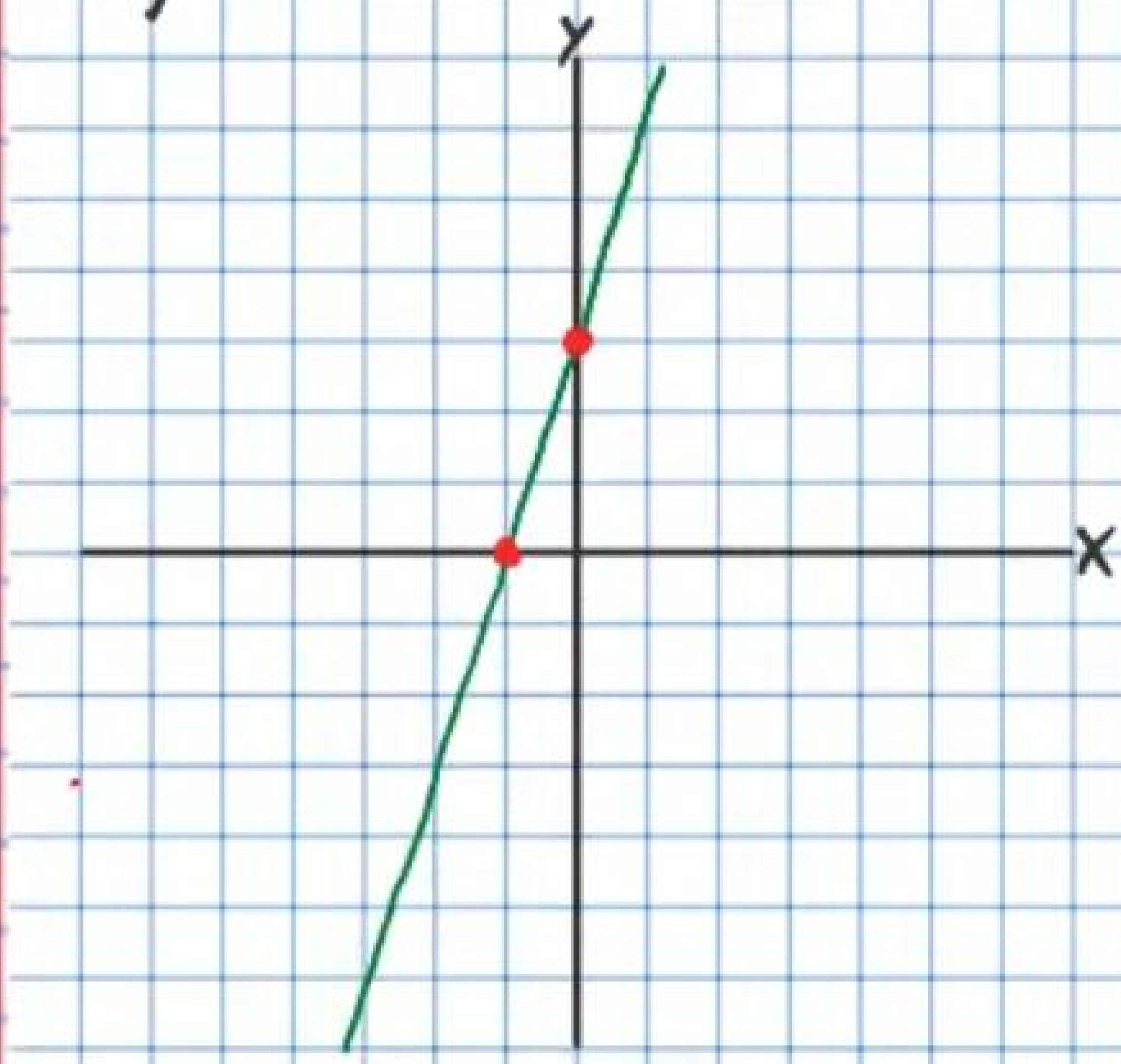


* Figure not drawn to scale.

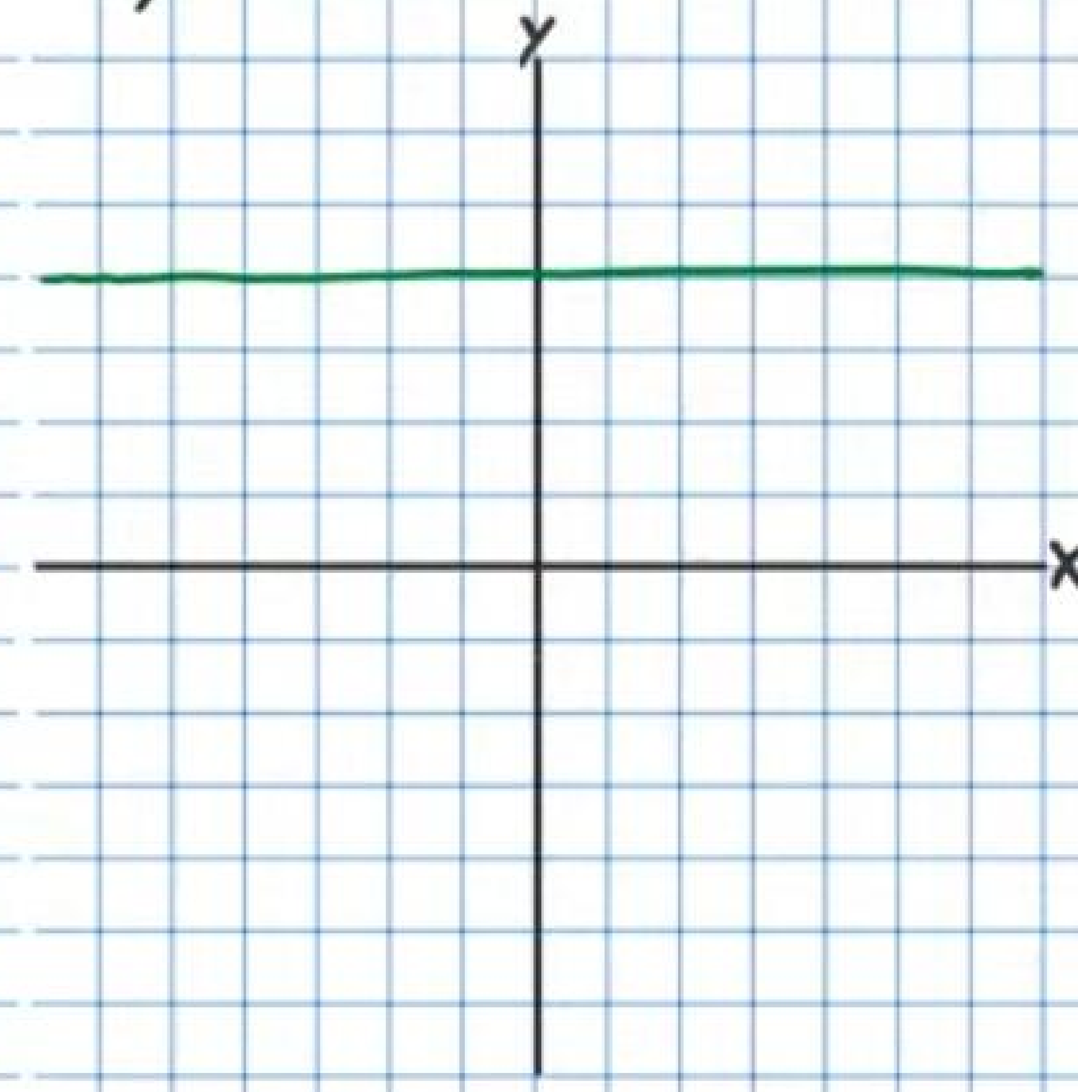
For each graph below, write an equation of the line shown.

(23)

a)

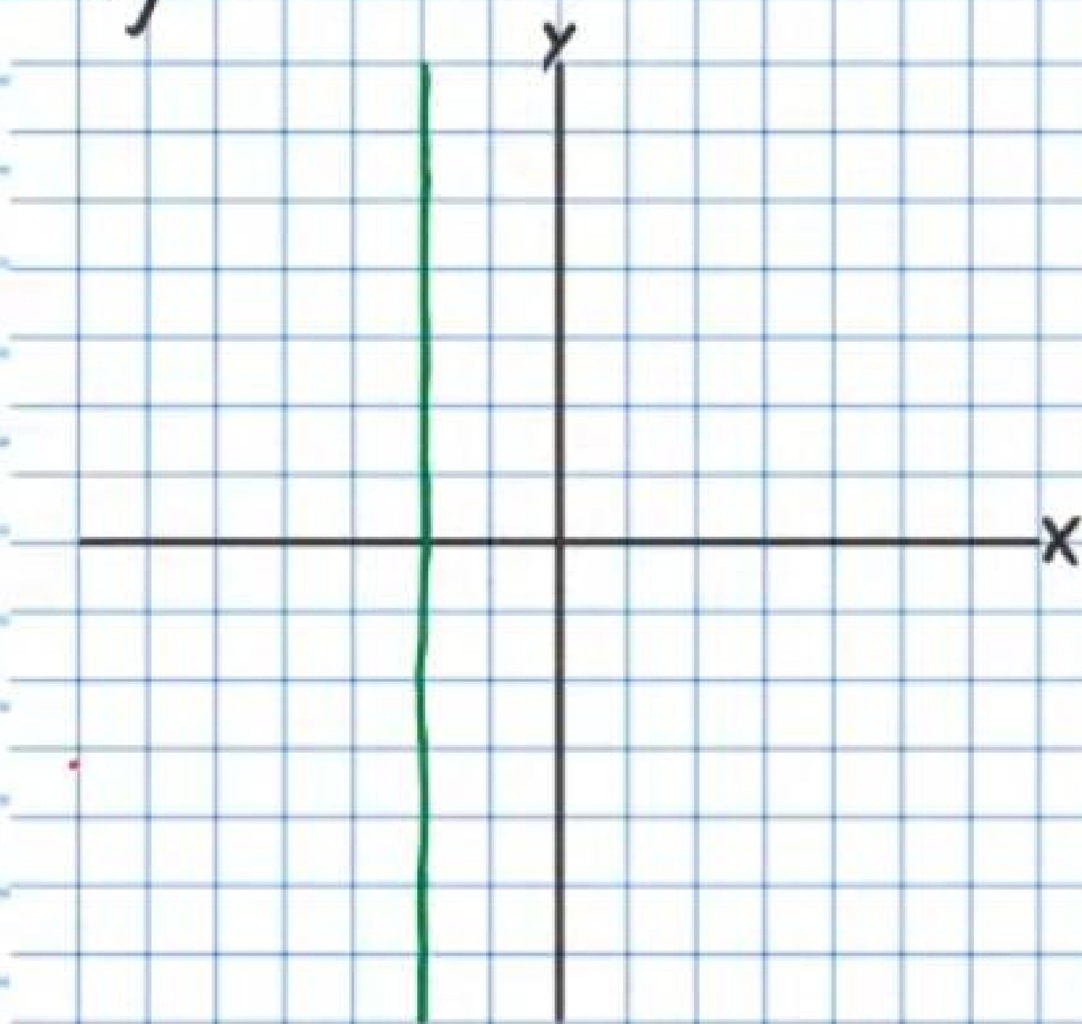


b)

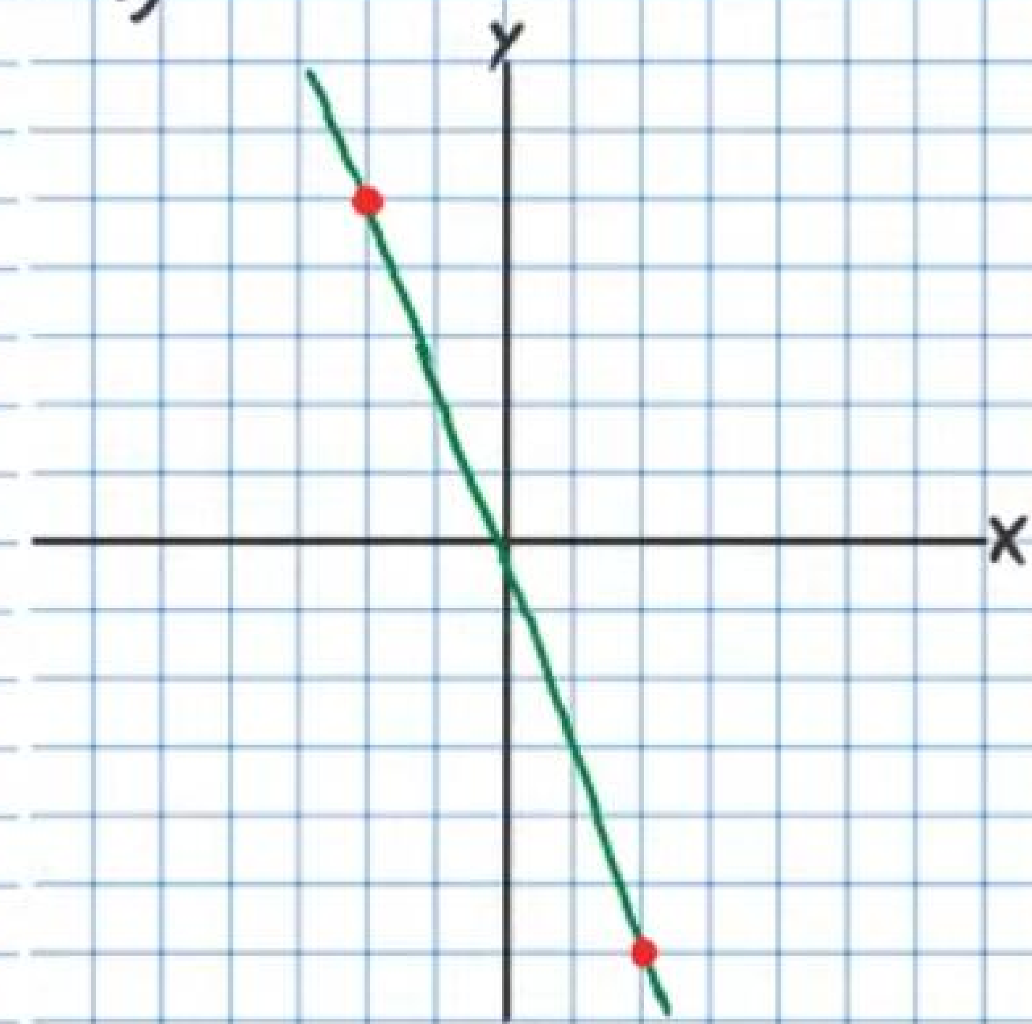


(24)

a)



b)



* You must use graphing paper for problem # 25.

25 a) Solve the system of equations below using the graphing method. As usual, label axes, intervals, and any points you use to solve the system.

$$\begin{cases} 4x + 5y = -2 \\ 3x + 8y = 7 \end{cases}$$

b) Then check your solution by solving the system algebraically (i.e. with substitution or elimination).

Beginning Algebra Test IV Answers

① $\boxed{x=-8}$, $\boxed{y=-23}$

② $\boxed{x=-4/17}$ $\boxed{y=1/17}$

③ $\boxed{x=2}$ $\boxed{y=3}$

④ $\boxed{x=1/2}$ $\boxed{y=-1/2}$

⑤ a) $\boxed{S}$ b) $\boxed{PS}$ c) $\boxed{SI}$ d) $\boxed{S}$

⑥ a) $\boxed{-11/3}$ b) $\boxed{y-3=-11/3(x-4)}$ or $\boxed{y+8=-11/3(x-7)}$
c) $\boxed{y=-11/3x+\frac{53}{3}}$ d) y-intercept $=\boxed{\frac{53}{3}}$ or $\boxed{17\frac{2}{3}}$

⑦ a) $\boxed{y=2x+11}$ b) $\boxed{y=\frac{5}{4}x-5}$

⑧ a) $\boxed{y=-x+4}$ or $\boxed{y-2=-(x-2)}$
b) $\boxed{y=-\frac{8}{9}x-\frac{5}{3}}$ or $\boxed{y-1=-\frac{8}{9}(x+3)}$
c) $\boxed{y=-\frac{3}{2}x-\frac{33}{2}}$ or $\boxed{y+9=-\frac{3}{2}(x+5)}$
d) $\boxed{y=\frac{1}{5}x-\frac{11}{5}}$ or $\boxed{y+\frac{11}{5}=\frac{1}{5}(x-0)}$

⑨ a) $\boxed{y-4=-3(x+2)}$ or $\boxed{y=-3x-2}$
b) $\boxed{y+7=\frac{3}{4}(x-0)}$ or $\boxed{y=\frac{3}{4}x-7}$
c) $\boxed{y=\frac{1}{2}x-2}$ or $\boxed{y+2=\frac{1}{2}(x-0)}$
d) $\boxed{y=\frac{2}{3}x+4}$ $\boxed{y-0=\frac{2}{3}(x+6)}$

⑩ $\boxed{P=9,\ F=15}$

⑪ $\boxed{C=9}$, $\boxed{W=2}$

⑫ $\boxed{P=\$5.50}$ $\boxed{T=\$2.80}$

⑬ $\boxed{D=12}$ $\boxed{N=9}$

⑭ $\boxed{5\text{ ft}, 6\text{ ft}, 8\text{ ft}}$

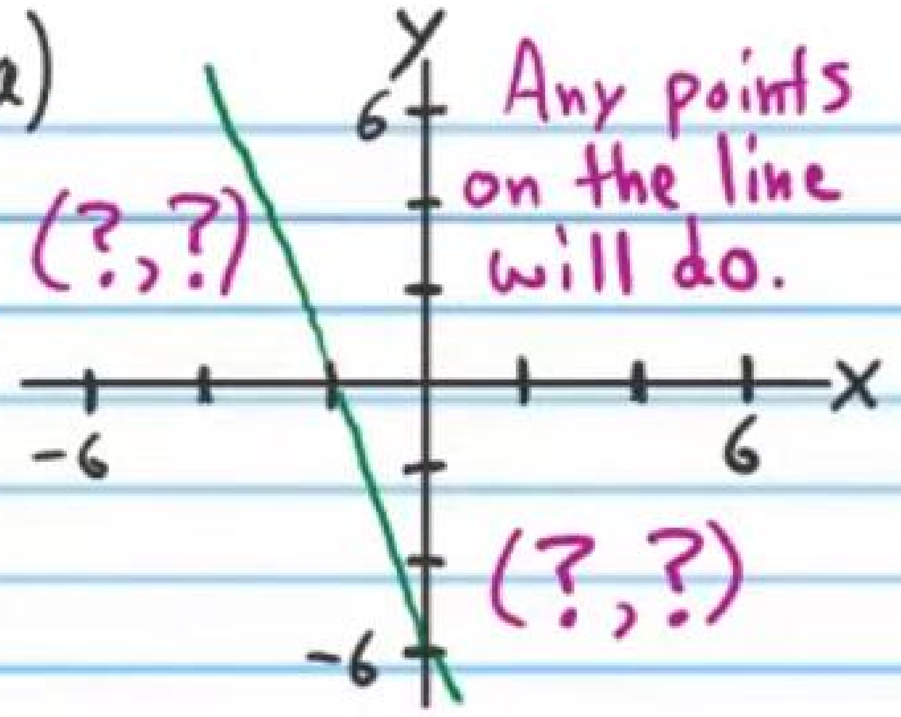
⑮ a) $\boxed{84\text{ ft}}$ b) $\boxed{\frac{\sqrt{15}}{2}\text{ seconds}}$ c) $\boxed{\sqrt{5}\text{ seconds}}$
d) $\boxed{5/2\text{ seconds}}$ or $\boxed{2.5\text{ seconds}}$

⑯ $\boxed{N=7.5\text{ kg}}$ $\boxed{C=12.5\text{ kg}}$

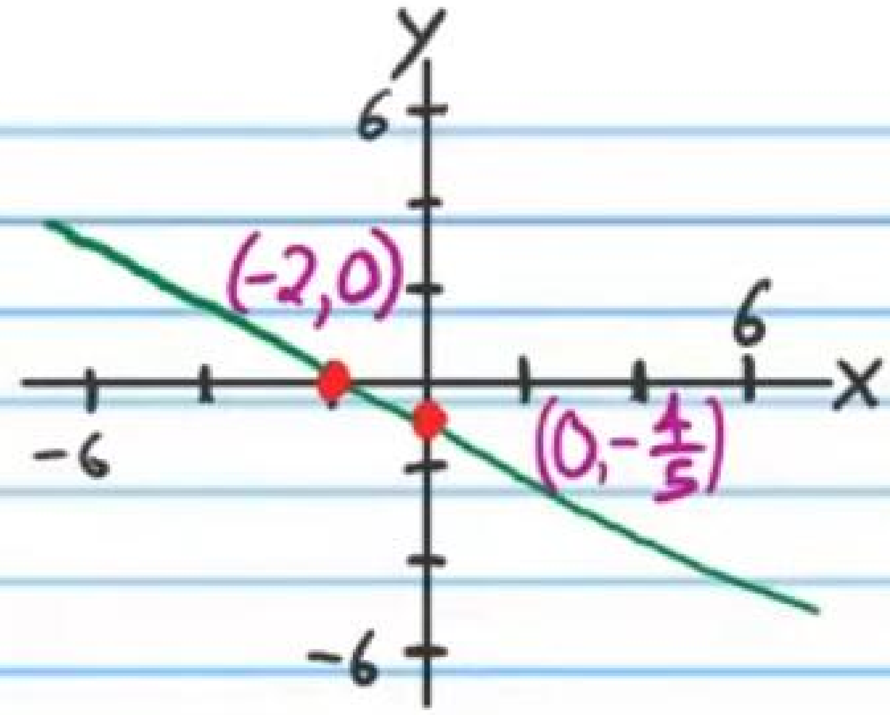
⑰ $\boxed{V_9=7.5\text{ oz}}$ $\boxed{V_1=4.5\text{ oz}}$

(18) a)

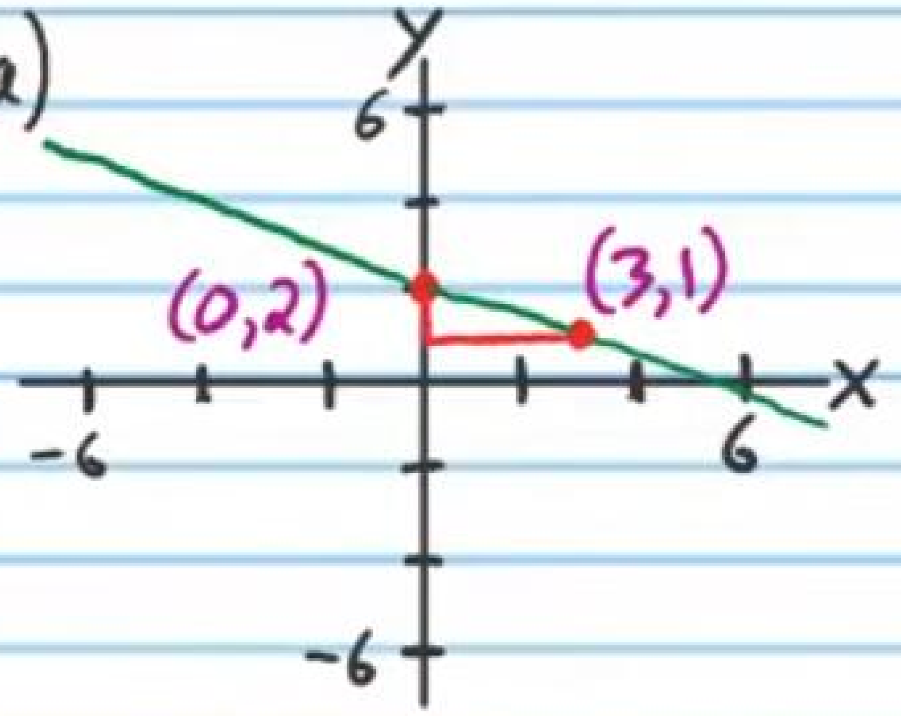
Any points on the line will do.



b)



(19) a)



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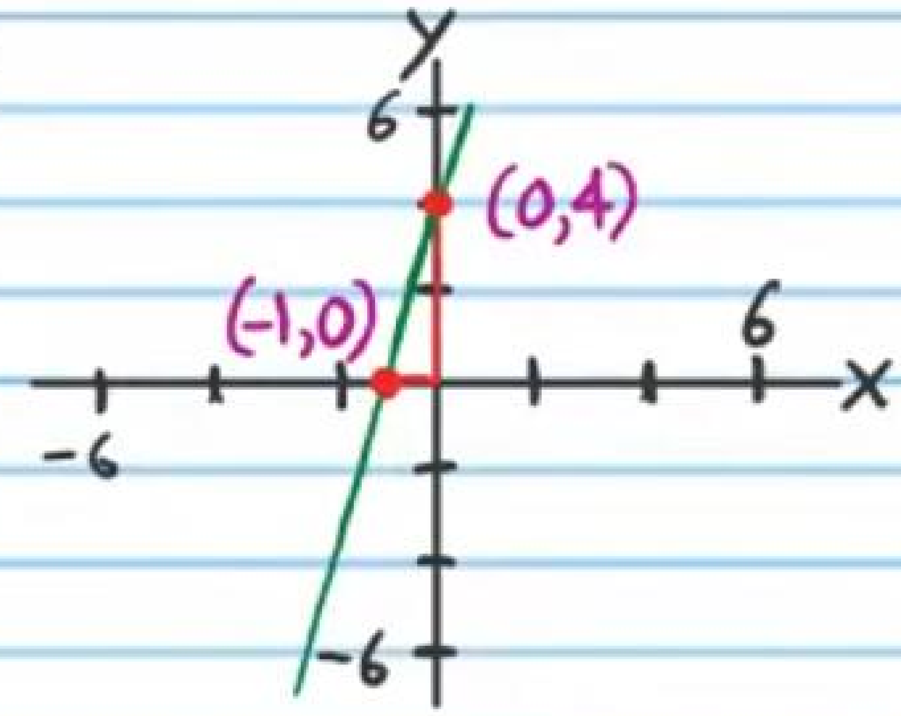
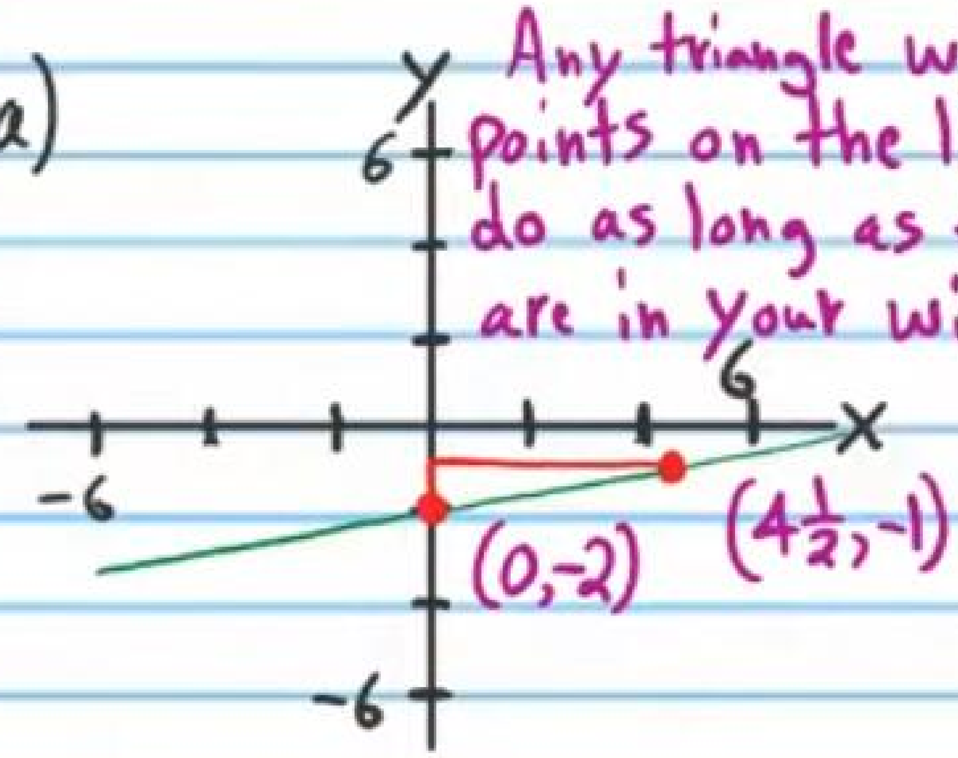


(20) a)

Any triangle with points on the line will do as long as the points are in your window.



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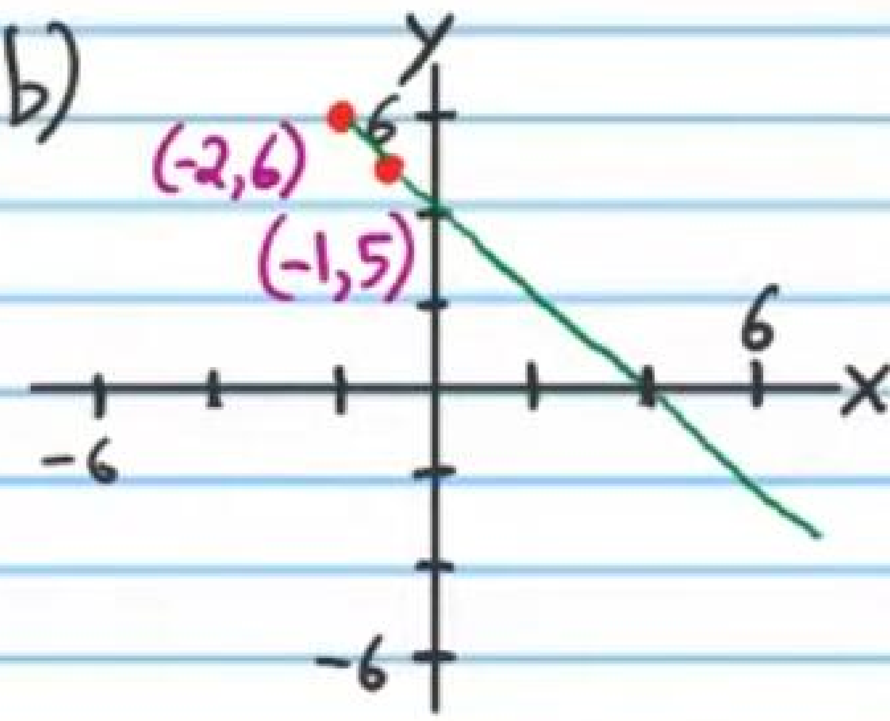
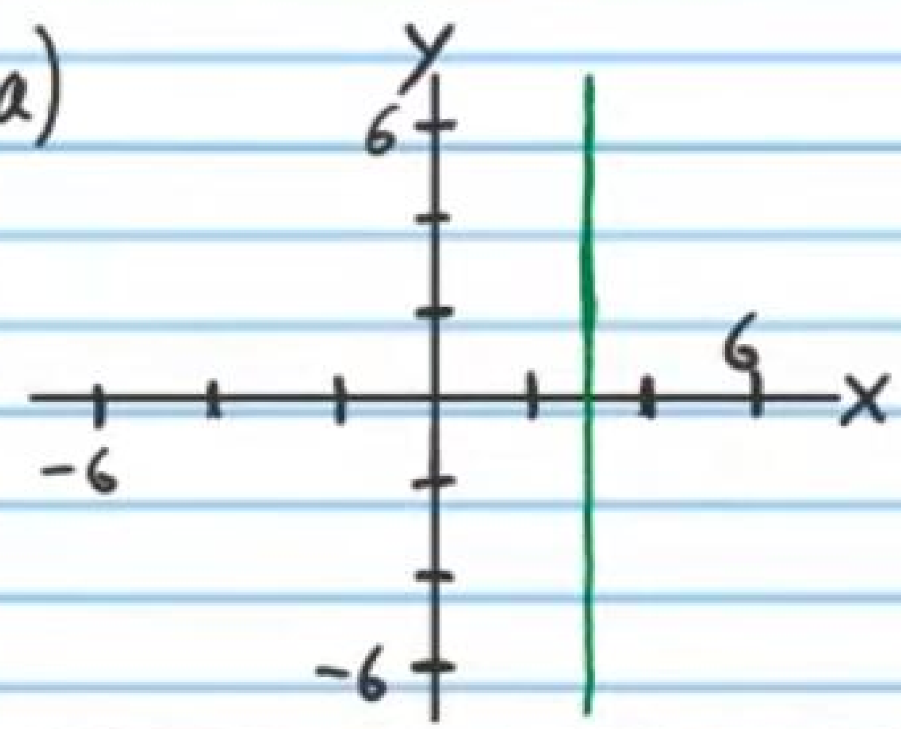


(21) a)



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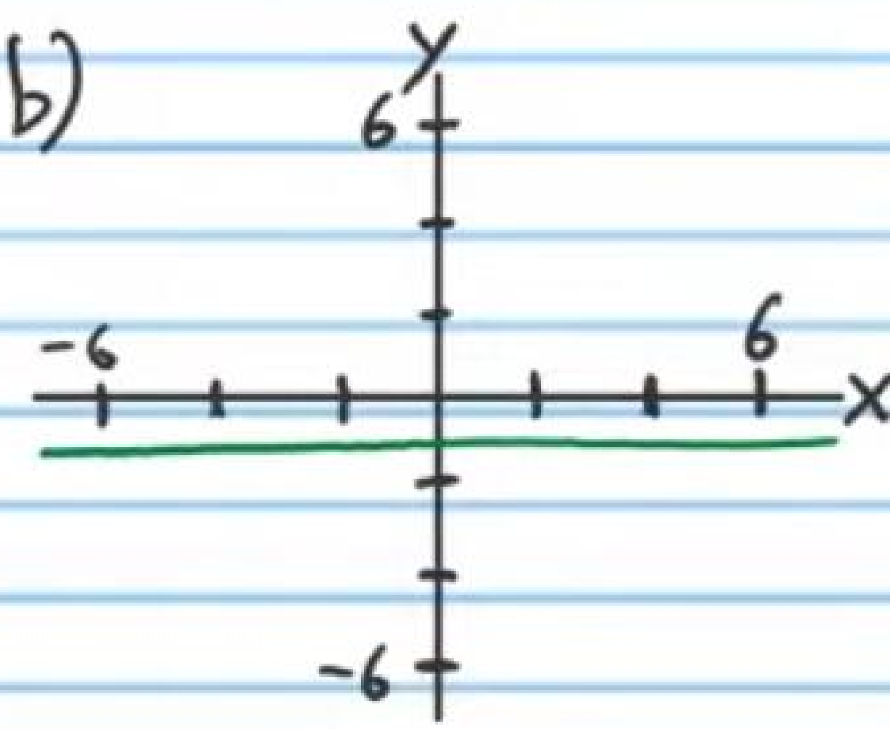


(22) $\boxed{W = 14\text{ m}}$ $\boxed{L = 10\text{ m}}$

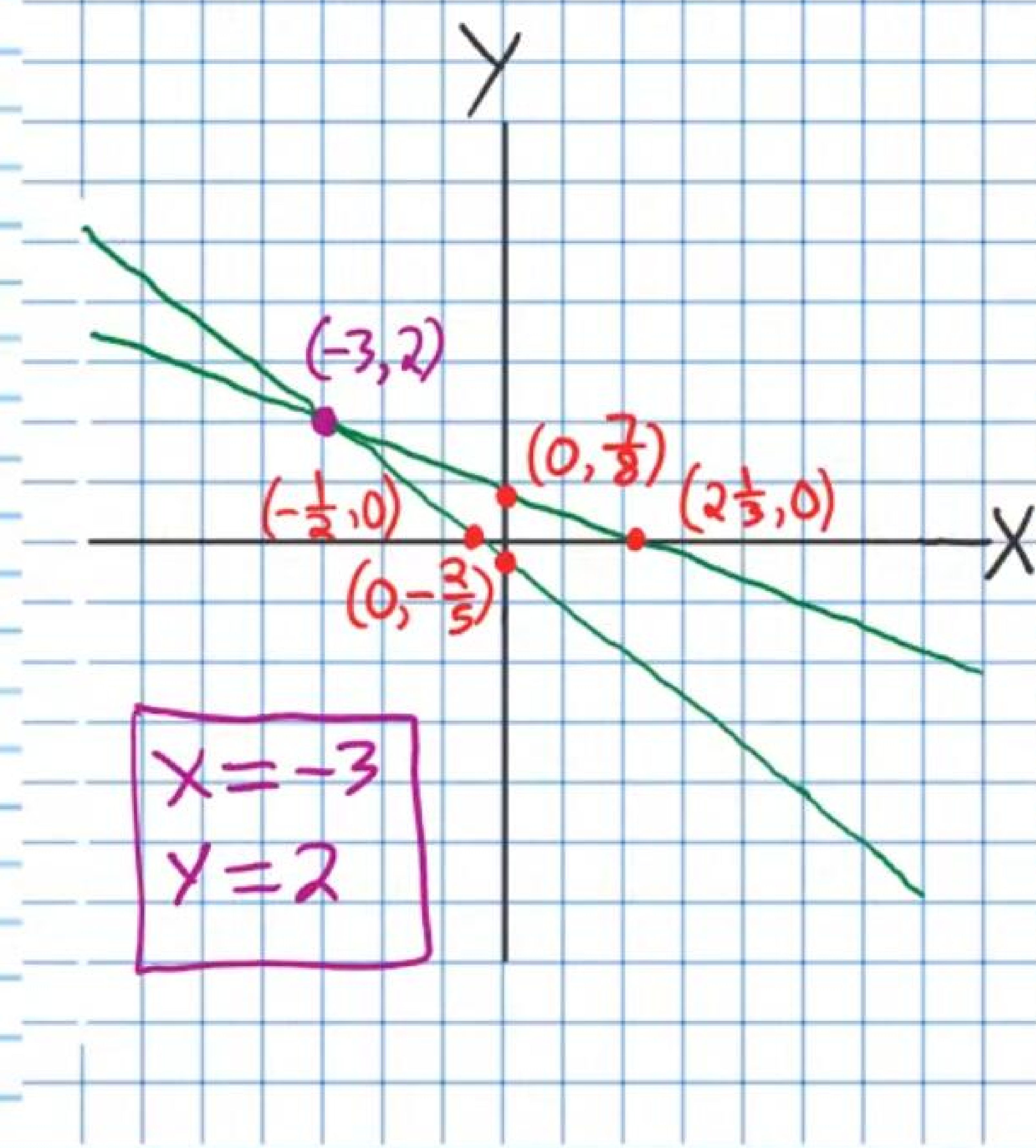
(23) a) $\boxed{y = 3x+3}$ or $\boxed{y-3 = 3(x-0)}$ or $\boxed{y-0 = 3(x+1)}$

b) $\boxed{y = 4}$

(24) a) $\boxed{x = -2}$ b) $\boxed{y = -\frac{11}{4}x - \frac{1}{2}}$ or $\boxed{y+6 = -\frac{11}{4}(x-2)}$ or $\boxed{y-5 = -\frac{11}{4}(x+2)}$

(25) a)

★ Other points will do as long as they are on the lines.



b) Give yourself two points if your work is done properly and it produces $x = -3$ and $y = 2$.